

Topics of Resilience: Improving Historical Phase Recognition with Paleo Climate, Risk Management and Topic Modeling

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Abstract

This paper presents a pilot study framing societal stability amid climate change as a Historical Phase Recognition task. Using topic modeling and portfolio risk management, we analyze the interplay between short textual descriptions of historical events and paleo-climatic data. By applying XGboost to these multimodal features, we achieve a peak balanced accuracy of 0.642, outperforming the State-of-the-Art (0.553) in detecting historical phases of rise and decline. The results demonstrate that, while LLM-driven topic modeling provides a solid baseline, additional features which can capture non-linear temporal trends can improve classification performance.

Keywords

Cultural Analytics, Climate Change, Historical Phase Recognition, Topic Modeling

1. Introduction and Related Work

Understanding long-term patterns of social responses to climate variability is crucial for strategic decision-making about contemporary challenges involving climate change [1]. We address this problem as a Historical Phase Recognition task [2], aligning paleo-climatic data [3] and textual descriptions of events in historical societies, categorized by structural-demographic cycles [4]. We apply Portfolio Risk Management (PRM) and Topic Modeling (TM) by means of Large Language Models (LLMs), in order to extract features from data and test how much the climatic data, treated as a risk factor, helps the classification of historical phases from text.

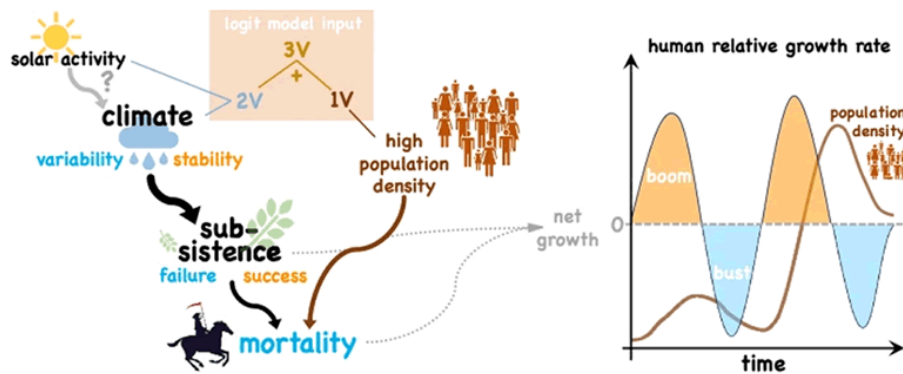


Figure 1: Influence of climate changes on human subsistence. Adapted from Wirtz et Al. 2024 [3]. License: <https://creativecommons.org/licenses/by/4.0/>

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As depicted in Figure 1, the interplay between climatic variability and human subsistence is a fundamental driver of social resilience. Solar activity, as a primary forcing mechanism, significantly influences climate stability, which in turn dictates the success or failure of various subsistence strategies. These environmental fluctuations directly modulate demographic dynamics. Human population growth is characterized by cyclical boom and bust phases, where mortality rates are often a function of both population density and resource availability. During boom periods, high climatic stability promotes subsistence success, facilitating rapid and positive relative growth rates of population. Conversely, during bust periods, increased climatic volatility triggers subsistence failures. These failures exacerbate mortality through mechanisms such as malnutrition and resource competition, forcing societies to adapt, or collapse. Dynamics such as resource competition, cultural adaptation and societal collapses are described by the Structural-Demographic Theory (SDT) [5] as cycles [6]. The structural-demographic cycle, depicted in Figure 2 is a theoretical historical pattern of societal change that lasts about 100 years, driven by the interplay between population, elites, the State, and the competition for power positions [7].

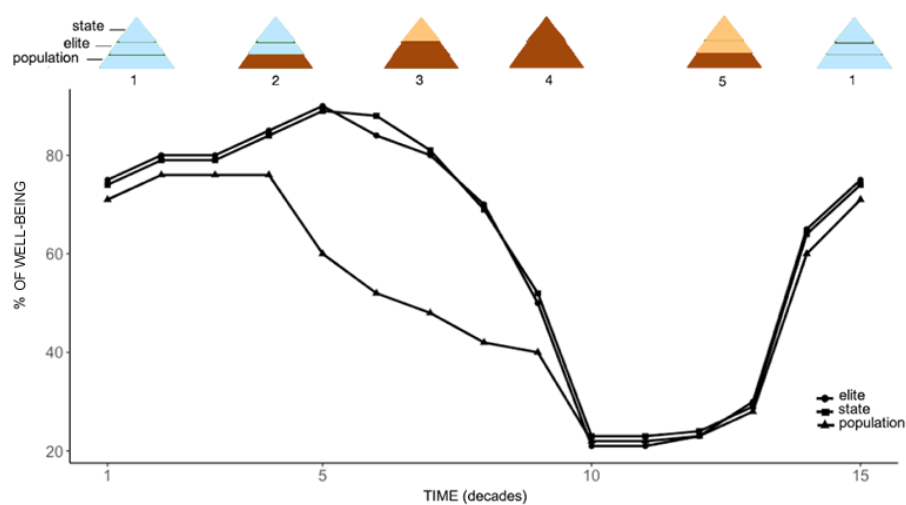


Figure 2: Schema of the Secular cycles phases according to the Structural Demographic Theory. Adapted from Hoyer et Al. 2025 [8] changing colors and terminology. License: <https://creativecommons.org/licenses/by/4.0/>

Structural-demographic cycles start with an expansion phase (1), characterized by increasing social cohesion and State control over the population, often followed by demographic growth. The Elizabethan Age in England (1550s), the Roman principate under Augustus and post-war Italy (1950s) exemplify this phase. This eventually transitions into a population immiseration phase (2), where population growth outpaces resources. Examples of this phase are the Qing Dynasty in the early 19th century, the United States during the 1970s and Italy during the 1980s. Population growth leads to an elite overproduction phase (3) with falling wages, and a bloated elite class fighting for a fixed number of power positions. Italy and the United States in the 2000s are examples of this phase. This tension culminates in a State stress phase (4), marked by state fiscal collapse, intense elite infighting, and widespread civil unrest or revolution. France in the 1780s and Italy during the 2010s are examples of this phase. Finally, a crisis phase (5) marks an intercycle period, where population or elite numbers are drastically reduced through conflict, eventually resetting the social carrying capacity and clearing the path for a new cycle. Examples include France in the 1790s, Italy in the 1940s, and the US during the Civil War in the 1860s and the New Deal era in the 1930s. The five phases can be grouped into rising (1,2) and declining (3,4,5) parts of the cycle [9] to perform binary classification.

The use of portfolio management techniques, originally developed for financial markets to balance risk and return, has increasingly been adopted in the fields of cultural evolution and climate change. In particular, research in climate adaptation applied portfolio strategies to optimize water management decision making in Australia [10], and adopted metrics such as Value at Risk [11] to model how societies and ecosystems survive under conditions of extreme uncertainty. Recently, climate change researchers

utilized PRM to mitigate the risk of excessive land-use demand by systematically comparing diverse combinations of carbon dioxide removal technologies to identify the most area-efficient and technically viable strategies for achieving global climate targets [12]. In cliodynamics, portfolio techniques help explain why certain social behaviors or institutional structures persist. For instance, a society might maintain both high-yield agriculture and low-yield, but drought-resistant, foraging. This is modeled as a risk-management strategy where the return is caloric stability and the risk is famine [13]. Current scientific literature in topic modeling marks a significant transition from traditional probabilistic models, like Latent Dirichlet Allocation (LDA), toward LLM-enhanced and LLM-exclusive topic extraction. Consensus is growing about the fact that LLMs solve the long-standing interpretability gap via pre-training [14] by providing human-readable topics and handling semantic nuances that bag-of-words models miss. A portion of the topic modeling literature is dedicated to proving that LLMs are viable alternatives to the 20-year dominance of LDA [15] but this requires new evaluation metrics, because traditional measures like Coherence Scores were designed for word-clusters. Recent research shows that LLM-generated topics have Coherence Scores closely aligned with Normalized Pointwise Mutual Information (NPMI), a metric known for its strong correlation with human judgment [16], suitable for the evaluation of LLM-extracted topics.

Historical Phase Recognition is a novel computational task aimed at recognizing historical phases, like the ones described by the structural-demographic cycles, from text or other sources. Previous attempts at this task revealed that transformers and event extraction applied to textual descriptions of historical decades perform poorly [2], reaching a peak balanced accuracy of 0.553 in the binary classification of "rise" and "decline" classes. Here we add paleo climatic and PRM features, and exploit a LLM to extract high-level topics from text. The overall goal of this study is twofold: first, we aim to improve the performance of Historical Phase Recognition; second, we explore the relationship between societal change (topics) and climate (paleoclimate temperature data). Specifically, we seek to answer two research questions (RQs):

- RQ1: to what extent climate-based PRM features and LLM-based topics help improving Historical Phase Recognition?
- RQ2: are LLM-based topics useful for gaining insight into responses to climate change?

Responses to these research questions will allow us to better understand topics and contexts of climate-related decline in past societies, and gain insights about possible responses. The paper is structured as follows: in Section 2 we describe the dataset, in Section 3 we define the methodology, then we report the results of the experiments in Section 4 and we draw our conclusion in Section 5.

2. Data

There are not many corpora available to explore historical societies and paleo-climate using computational linguistics techniques. Among the corpora about past societies there are: the Seshat project [17, 18], that released different datasets about past societies, some of which include textual description of past societies at the level of centuries; the GINI (Global dynamics of INequality) project released some data on past societies with a high time resolution [19] but does not provide textual descriptions of societies and events. The Chronos dataset [4], which is derived from Seshat, provides descriptions of events about past societies at the level of decade, and is the first to be annotated with SDT labels. Among the available corpora on paleo-climate there are: PaleoClim [20], providing non-continuous climate data for key time slices with very high resolution (intra-annual); the Global Holocene Paleotemperature Database [21] contains continuous multi-proxy time series over the last 12,000 years with a mixed time resolution (decadal to centennial); the Phanerozoic Climate Simulation Dataset [22] presents a continuous, model-based dataset simulating global climate dynamics across the entire Phanerozoic Eon, but with a multi-millennial time resolution. The Multicentennial Paleo Climatic dataset [3] combines radiocarbon archaeological proxy records (estimating historical regional population shifts) alongside multi-proxy paleoclimate data linking long-term climate dynamics to human socio-demographic history at the time resolution of decades.

The dataset assembled for this study, called "Chronos-Climate dataset", aligns subsets of Chronos and the Multicentennial climatic database at the level of decade, and was selected for the compatible time resolution and depth. This dataset provides a multi-dimensional perspective on historical evolution of societies and climate. In particular, we aligned textual descriptions of historical societies with climatic data by decade and geographic area, keeping only those decades having a textual description in the Chronos dataset.

Table 1

Structure and variables of the integrated dataset.

feature	description	source
decade	The temporal point of the observation.	
society	Geographical/Societal identifier (e.g., <i>eg_naqada_1</i> , <i>mx_aztec_emp</i>).	[17]
events description	Textual description of historical events and transitions.	[4]
sdcyclephase	Structural-Demographic phase (1–5).	[4]
relative growth rate	Standardized demographic growth trajectory.	[3]
paleotemp last interpolated	Standardized paleo temperature, interpolated by last value.	[3]

The structure of this integrated dataset is reported in Table 1. The dataset consists of 534 observations across 72 distinct historical societies, spanning a temporal range from approximately 6600 BC to 1500 AD. The Structural-Demographic categorization of historical decades is annotated on a subset of about 100 exmples, by two human raters, who were not experts in history, trained on the SDT, who obtained a Fleiss' Kappa [23] of 0.455 as inter-rater agreement. One of the two then annotated the entire corpus. Attaining strong agreement is inherently complex in this task, due to the temporal constraints of SDT phase labeling. A single phase misalignment early on can propagate across successive time-steps, resulting in a compounding decline in overall agreement [24]. For this reason, we utilized a standard secular cycle model as initial annotation schema, and we asked the raters to express their judgment by modifying the schema according to the following guidelines, without adjudication of disagreements:

1. Read the textual description to identify key events: wars, reforms, rulers, population, elites, disasters, epidemics, alliances or treaties, socio-economic context, famines or financial stress, protests or movements, religions.
2. Use polity identifiers (already present in the dataset, derived from expert annotation done in the Seshat project[17]) to find the start and end decades of cultures. The end of a culture probably marks a crisis period.
3. Starting from the beginning of a society, initially assign the sequence of labels of a standard secular cycle model: 1,1,2,2,3,3,4,4,4,5 and then evaluate whether to keep or change the labels in each decade. It is possible to have longer or shorter cycles. There can be only one label 5 (crisis) per cycle. A polity can have one or more cycles.
4. Select one of the following labels to describe the decade reported in the textual description: 1=growth. A society is generally poor when it experiences renewal or change followed by demographic (but not always territorial or economic) growth. Reforms, alliances, wars won or similar events are potential indicators of this phase. 2=impoverishment of the population. Potential economic and/or territorial expansion slows while demography continues to expand. The elite takes much of the wealth and defines the status symbols. Stability and external attacks are potential indicators of this phase. 3=Overproduction of the elites. The wealthy seek to translate their wealth into positions of authority and prestige. The population becomes poor. Movements, protests, and wars are potential indicators of this phase. 4=State stress. The elites want to institutionalize their advantages in the form of low taxes and privileges that lead the state into fiscal difficulties. Wars, protests and changes in the elite are potential indicators of this phase. 5=Crisis. a triggering event such as a war, revolt, famine or disaster that the state is unable to manage leads to a new configuration of society. Emigration of elites, subjugation to other societies, civil wars or profound reforms are potential indicators of this phase.

5. Use the progressive order of the phases if no textual description is available for the decade.
6. Make sure there is a progressive order of the labels (e.g. phase 3 must follow phase 2). All labels can be repeated in the following decade except the crisis phase, which conventionally lasts one decade.

Despite using the standard secular cycle model, agreement remained moderate because phase cues are inherently ambiguous and context-dependent. For example wars can signal both expansions, typical of phases 1 and 2, or decline, typical of phase 4. The same holds for revolts, that can occur during population immiseration phases, as well as towards the end of the cycles, in phases 4 and 5. Interpreting many cues in context, like internal and external situation of the polities, is crucial for this annotation task. The relative growth rate reports global human population dynamics in one metric, using summed probability distributions of radiocarbon dates ($> 91,000$ dates) with multicentennial growth cycles synchronized across continents, driven by climate stability and solar activity. The "paleotemp last interpolated" feature provides a localized paleo temperature proxy, interpolated by the last value recorded. The following section describes how we used the data in the experimental settings.

3. Methodology

This study proposes a novel analysis framework that integrates PRM [25, 26] and TM [27, 28]. We consider societies in the Chronos-Climate dataset as portfolios with two assets: paleo temperature variation and relative growth rate of the population. Paleo temperature represents the climatic capital of a society and relative growth rate represents its demographic capital. Both PRM and TM are used as feature extraction techniques. To apply PRM, we preprocess the Chronos-Climate dataset with asset normalization. Relative Growth rate and Paleo Temperature are standardized to a common scale to facilitate cross-asset aggregation and ensure that disparate data types, such as temperature anomalies and demographic shifts, are comparable. PRM analysis provides the following new features:

- Return on paleo temperature
- Return on relative growth rate
- Returns
- Portfolio return
- Portfolio risk
- Portfolio value at risk.

For each decade, Returns reports the net changes in paleo temperature, relative growth rate, and a combination of the two. Positive returns signify a holistic improvement in the combined socio-climatic-demographic state, in other words: success in subsistence. Portfolio return reports the average improvement at the level of society: portfolio risk, reports the volatility of subsistence of a society, calculated as the standard deviation of Portfolio return across the society's specific timeline. Value at Risk (at 95% confidence interval) is calculated at societal level as well and represents the expected worst-case decline of a society in a time unit (decade), acting as a proxy for the severity of potential systemic collapses. We extracted topics from the event descriptions using the following prompt on Gemini:

act as an expert of history and climate change. Extract up to 3 high level topics as single words, such as WAR, MIGRATION, EXPANSION, DISASTER etc, from the following text: *events description*

Figure 3: Prompt for the extraction of topics from texts.

We evaluated the quality of the topics extracted with the Normalized Pointwise Mutual Information (NPMI) between the events description and the topics extracted. We tokenized the text of events description (removing punctuation and common stopwords), and compared it against the individual topic labels (e.g., "WAR", "POLITICS"). Then we calculated the probability of occurrence for each topic

$P(t)$, each descriptive word $P(w)$, and their joint occurrence in the same row $P(t, w)$. The NPMI for each pair was calculated as:

$$NPMI(t, w) = \frac{\ln \left(\frac{P(t, w)}{P(t)P(w)} \right)}{-\ln(P(t, w))}$$

where the logarithm (ln) is used for calculating the "surprise" factor, in other words how much more often the word and topic appear together than we would expect by pure chance. If the ratio is greater than 1, ln is positive (the word is associated with the topic), if the ratio is less than 1, ln is negative (the word and topic avoid each other). The average NPMI of the topics extracted is 0.271, indicating good semantic associations between the descriptive text and the assigned topics. Finally, we binarized the Structural-Demographic cycle labels by grouping phases 1 and 2 into a 'rise' class and phases 3, 4, and 5 into a 'decline' class. The resulting distribution is relatively balanced, with 'rise' accounting for 42.9% of the data and 'decline' for 57.1%. The final dataset is freely available on Huggingface¹.

4. Experiments and Discussion

We addressed Historical Phase Recognition as a binary task where we exploit PRM features and topics to classify "rise" and "decline" with transparent algorithms: linear (logistic regression, linear discriminant analysis) and non-linear (decision trees, XGboost). PRM are numeric and can be used for classification without any preprocess. Instead, we treat topics as unigrams and, in order to reduce sparseness, we keep only the 100 most frequent topics. We evaluate the performance of the classification task with precision, recall, weighted-averaged F1 and balanced accuracy, on a 70-30% training-test split. We also rank feature importances and model coefficients to retrieve the best performing feature for each trial.

Table 2

Results of the classification experiments. We used logistic regression (lr), linear discriminant analysis (lda), decision trees (dt) and XGboost (xgb) to classify "rise" and "decline" classes from risk and topic features. We compute the performance difference against topics (Δ bal acc) and marked best performance in bold.

algorithm	features	P	R	w-avg F1	bal acc	Δ bal acc	best feature
lr	topics	0.63	0.62	0.62	0.626	-	ELITE
lda	topics	0.55	0.54	0.54	0.549	-	FORTIFICATION
dt	topics	0.61	0.61	0.61	0.608	-	WAR
xgb	topics	0.63	0.63	0.63	0.629	-	PROSPERITY
lr	PRM	0.23	0.48	0.31	0.500	-	return on growth rate
lda	PRM	0.23	0.48	0.31	0.500	-	return on growth rate
dt	PRM	0.54	0.54	0.54	0.540	-	paleo temperature
xgb	PRM	0.57	0.57	0.55	0.561	-	return on growth rate
lr	both	0.57	0.55	0.53	0.557	-0.069	INNOVATION
lda	both	0.59	0.57	0.56	0.575	0.026	FORTIFICATION
dt	both	0.63	0.62	0.60	0.611	0.003	paleo temperature
xgb	both	0.68	0.65	0.63	0.642	0.013	PROSPERITY

Results, reported in Table 2, indicate that topic-based features provide a solid baseline, but the integration of PRM-based features is essential for peak model performance. With the exception of logistic regression, models generally performed best in our experiments when using both topics and PRM features. Moreover, PRM features only work with non-linear models, while topics alone yield good performances with both linear and non-linear models. The best performing algorithm is XGboost, with a balanced accuracy of 0.642 and a weighted-average F1 of 0.63. This is a clear improvement over the State-of-the-Art baseline for this task: 0.553 balanced accuracy [2] obtained with linear discriminant analysis paired with BERT features. The effectiveness of XGboost classifier in this context stems from its ability to efficiently generalize over sparse data and control model complexity, reducing overfitting

¹<https://huggingface.co/datasets/facells/chronos-climate>

even on such a relatively small dataset. To control for dataset size, we repeated the experiment on the Chronos-benchmark² [2] (1422 records) by extracting LLM-based topics from English text for binary classification. Using XGBoost, we achieved results slightly superior to those of the Chronos-climate dataset: 0.77 precision, 0.64 recall, 0.68 F1, and 0.63 balanced accuracy. This demonstrates that LLM-based topic extraction is effective in Historical Phase Recognition. On Chronos-climate, Both logistic regression (lr) and linear discriminant analysis (lda) struggled significantly when using PRM features alone (precision as low as 0.23). In contrast, tree-based models (dt, xgb) handled these kinds of features much more effectively, suggesting that the relationship between climate, growth, risk and historical phase transitions is highly non-linear. The analysis of the best features reveals a fascinating mix of drivers. While linear models rely on topics like "ELITE", "FORTIFICATION" and "INNOVATION" to distinguish the two classes, the best predictors with non-linear models are more related to concepts like "PROSPERITY" and "WAR". It is interesting to note that decision trees, if allowed to select between topics and climate PRM-based features, choose paleo temperature as the root. To answer RQ1 (whether the PRM-based features help the classification): there is a general improvement using PRM-based features with non-linear models.

Table 3

Pearson correlations with P-value below 0.01 between topic frequencies and PRM-based climatic features.

features	ρ	P-value
SETTLEMENT and paleo temperature	0.172	0.000065
TRADE and paleo temperature	0.166	0.000120
PEACE and return on paleo temperature	0.134	0.001849
RITUAL and paleo temperature	0.133	0.002006
COMMUNITY and return on relative growth rate	0.122	0.004726
DISASTER and return on relative growth rate	0.120	0.005232
RESOURCE and return on relative growth rate	0.117	0.006829
TECHNOLOGY and paleo temperature	0.115	0.007590
TURMOIL and return on paleo temperature	0.111	0.009893
CIVILIZATION and paleo temperature	0.106	0.013714
ALLIANCE and return on paleo temperature	0.103	0.017351
STABILITY and paleo temperature	-0.105	0.014501
EVOLUTION and return on relative growth rate	-0.121	0.004864
EXPLOITATION and return	-0.131	0.002380
POLITICS and paleo temperature	-0.133	0.002019
EMPIRE and paleo temperature	-0.133	0.001965
CONFLICT and paleo temperature	-0.147	0.000638
EXPLOITATION and return on paleo temperature	-0.167	0.000102
REBELLION and paleo temperature	-0.175	0.000045
WAR and paleo temperature	-0.253	0.000000

To answer RQ2 (whether LLM-based topics are useful to get insights about responses to climate change) we extracted correlations between topics and PRM-based climatic features, reported in Table 3. The correlation analysis reveals a striking relationship between environmental variables and the socio-political dynamics of historical societies. Most notably, paleo temperature serves as a primary driver of both growth and strife; while warmer and stable climate is positively associated with constructive features like SETTLEMENT ($\rho = 0.172$) and TRADE ($\rho = 0.166$), there is also a profound negative correlation with systemic violence. In particular, WAR ($\rho = -0.253$) and REBELLION ($\rho = -0.175$) show the strongest inverse relationships with temperature, suggesting that cooling periods or climate instability historically acted as catalysts for conflict and domestic unrest. Furthermore, the data highlights the importance of returns on environmental and demographic factors. A society's ability to exploit climate stability, measured as the return on paleo temperature, is significantly linked to PEACE ($\rho = 0.134$) and ALLIANCE ($\rho = 0.103$), whereas the capacity to extract value from population growth

²<https://huggingface.co/datasets/facells/chronos-historical-sdt-benchmark>

(return on relative growth rate) correlates with management topics like COMMUNITY and RESOURCES. Collectively, these results, underscored by high statistical significance, suggest that LLM-based topic modeling is useful to extract meaningful and human-readable insights from complex data.

5. Conclusion

This study improved Historical Phase Recognition by integrating textual event descriptions with paleo-climatic and demographic data through the Chronos-Climate dataset. By treating societies as portfolios and applying Portfolio Risk Management (PRM) alongside LLM-driven topic modeling, we demonstrated that climatic stability is a fundamental driver of social resilience. Our results indicate that, while topic-based features provide a solid baseline, the integration of non-linear PRM features yield peak performance, with the XGboost algorithm outperforming the State-of-the-Art. Statistical analysis revealed that warmer, stable climates correlate with constructive societal traits like trade and settlement, whereas cooling and variable periods act as catalysts for systemic violence, such as war and rebellion. These findings validate the utility of LLM-based topics and multi-modal features in capturing complex, non-linear relations in historical data. Our results demonstrate that classification performance in Historical Phase Recognition from text can be improved significantly, and that this task could benefit from research in topic modeling to extract features. Future research in this task should focus on trying new models for target classes and combine different types of temporal data.

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Declaration on Generative AI

During the preparation of this work, the authors used GPT-4 and Gemini 3 in order to perform: Grammar and spelling check; Paraphrase and reword; Improve writing style. After using these tools/services, the authors reviewed and edited the content as needed and take full responsibility for the publication's content.

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